International Cybersecurity and Digital Forensics Academy (ICDFA)

Basic Computer Skills for Digital Forensics

COURSE CODE CIP-B101

Lab 1B: PC System Fundamentals, Disk Sectors and Partition Analysis

COURSE CODE WADF201

KALI

ABUBAKAR BELLO SADIQ

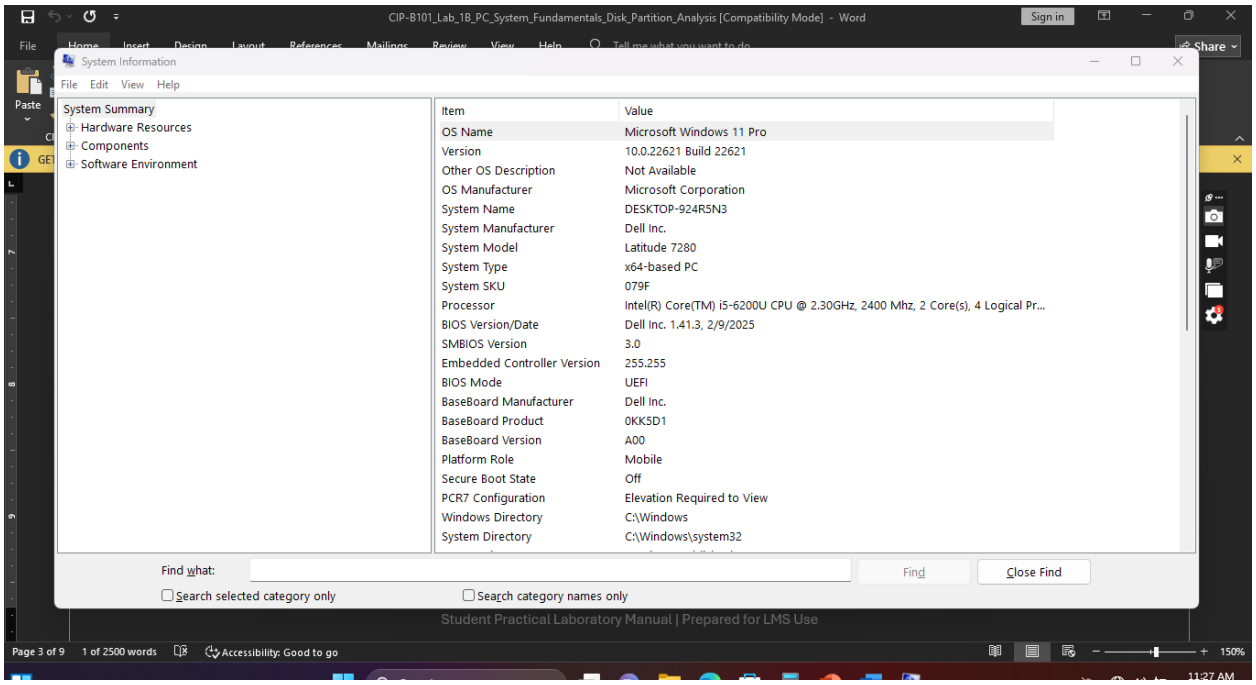
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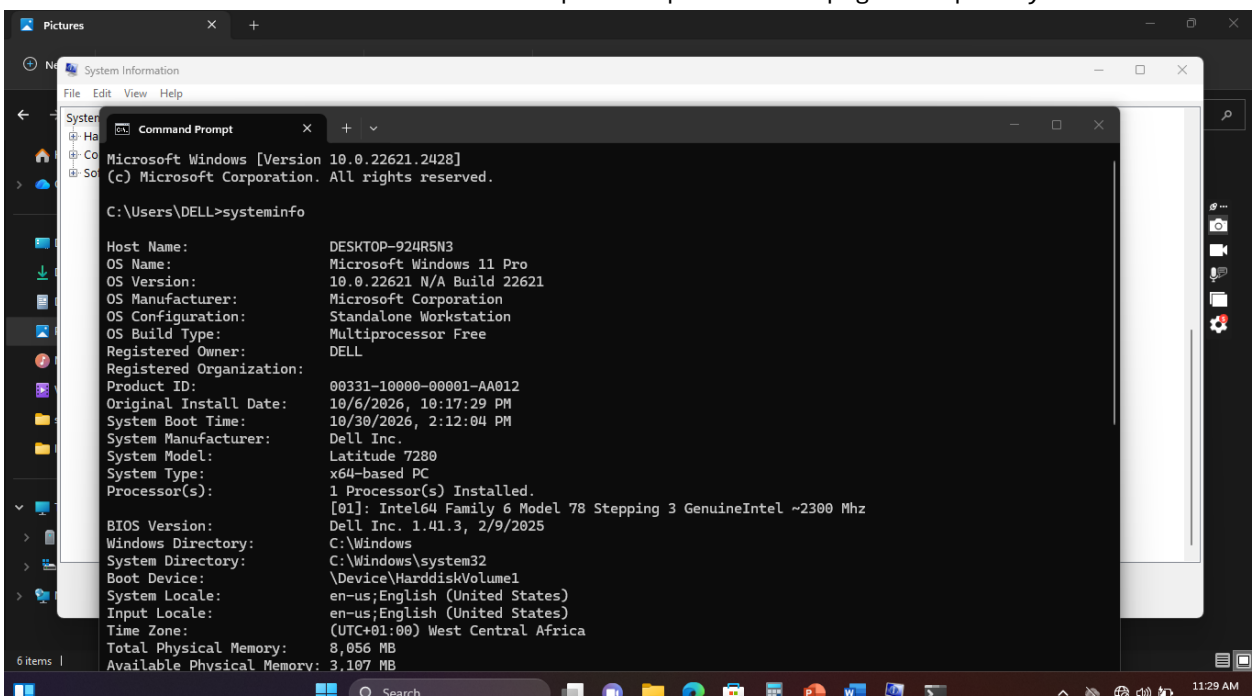
Part A - System Identification and Evidence Layer Mapping

A1. Capture basic system specification

- On Windows, open System Information by pressing Windows + R, typing msinfo32, and pressing Enter.

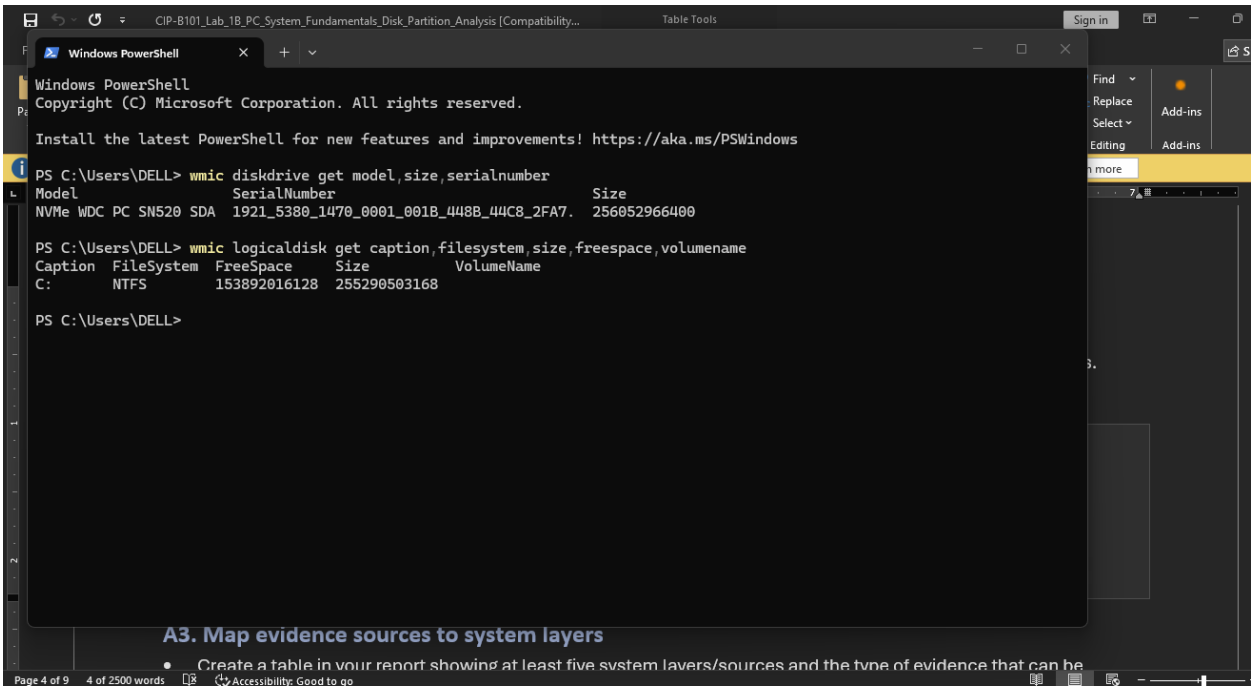


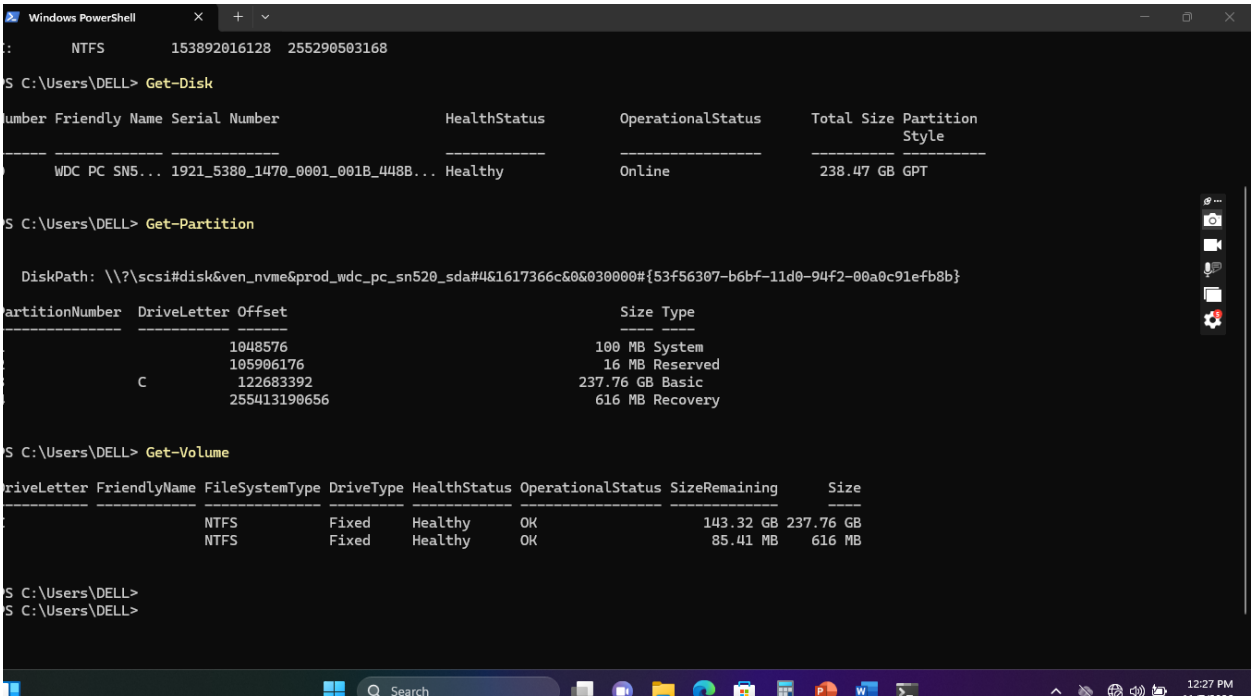
- Also run the command below in Command Prompt and capture the first page of output. `systeminfo`



A2. Capture storage information

- Open Command Prompt or PowerShell.


CIP-B101_Lab_1B_PC_System_Fundamentals_Disk_Partition_Analysis [Compatibility...
Table Tools
Sign in
Windows PowerShell
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.
Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows
PS C:\Users\DELL> wmic diskdrive get model,size,serialnumber
Model SerialNumber Size
NVMe WDC PC SN520 SDA 1921_5380_1470_0001_001B_448B_44C8_2FA7. 256052966400
PS C:\Users\DELL> wmic logicaldisk get caption,filesystem,size,freespace,volumename
Caption FileSystem FreeSpace Size VolumeName
C: NTFS 153892016128 255290503168
PS C:\Users\DELL>
Find
Replace
Select
Editing
Add-ins
Add-ins
A3. Map evidence sources to system layers
Create a table in your report showing at least five system layers/sources and the type of evidence that can be
Page 4 of 9
4 of 2500 words
Accessibility: Good to go


Windows PowerShell
: NTFS 153892016128 255290503168
S C:\Users\DELL> Get-Disk
umber Friendly Name Serial Number HealthStatus OperationalStatus Total Size Partition Style
WDC PC SN5... 1921_5380_1470_0001_001B_448B... Healthy Online 238.47 GB GPT
S C:\Users\DELL> Get-Partition
DiskPath: \\?\scsi#disk&ven_nvme&prod_wdc_pc_sn520_sda#4&1617366c&0&030000#{53f56307-b6bf-11d0-94f2-00a0c91efb8b}
artitionNumber DriveLetter Offset Size Type
1048576 100 MB System
105906176 16 MB Reserved
C 122683392 237.76 GB Basic
255413190656 616 MB Recovery
S C:\Users\DELL> Get-Volume
riveLetter FriendlyName FileSystemType DriveType HealthStatus OperationalStatus SizeRemaining Size
NTFS Fixed Healthy OK 143.32 GB 237.76 GB
NTFS Fixed Healthy OK 85.41 MB 616 MB
S C:\Users\DELL>
S C:\Users\DELL>
Search
12:27 PM

A3. Map evidence sources to system layers

The below table explain in details all the five layers and the types of evidence they generate, mapped to user and system activities. This explains in details system layers.

S/N	System Layer / Source	Description	Types of Evidence Generated
1	Applications	The software interfaces users interact with directly (e.g., web browsers, chat clients, email software).	Browser history, cache, cookies, chat logs, email headers/bodies, database files, and application-specific configurations.
2	Operating System (OS)	The software layer managing system resources and tracking user interactions with the environment.	Registry hives, event logs (login/logout events), link files (.lnk), Shellbags, Prefetch files, and account creation logs.
3	File System (Disk)	The structural organization of data on storage media (e.g., NTFS, ext4, APFS).	File metadata (MACB timestamps: Modified, Accessed, Created, MFT Born), deleted file remnants, slack space, and partition tables.
4	Volatile Memory (RAM)	The system's temporary, high-speed workspace that holds active data while the machine is powered on.	Running processes, active network connections, unencrypted passwords/encryption keys, open file handles, and clipboard data.
5	Network / Peripheral Layer	The hardware and logical interfaces connecting the system to external devices and networks.	MAC addresses, IP assignment logs (DHCP), connected USB device history (Vendor/Product IDs), and firewall/router traffic logs.

Explain why evidence may exist in more than one layer.

Why Evidence Exists in Multiple Layers, Digital evidence rarely stays confined to a single layer. Because of how modern computers process information, user actions ripple down through the infrastructure, leaving artifacts across multiple layers simultaneously.

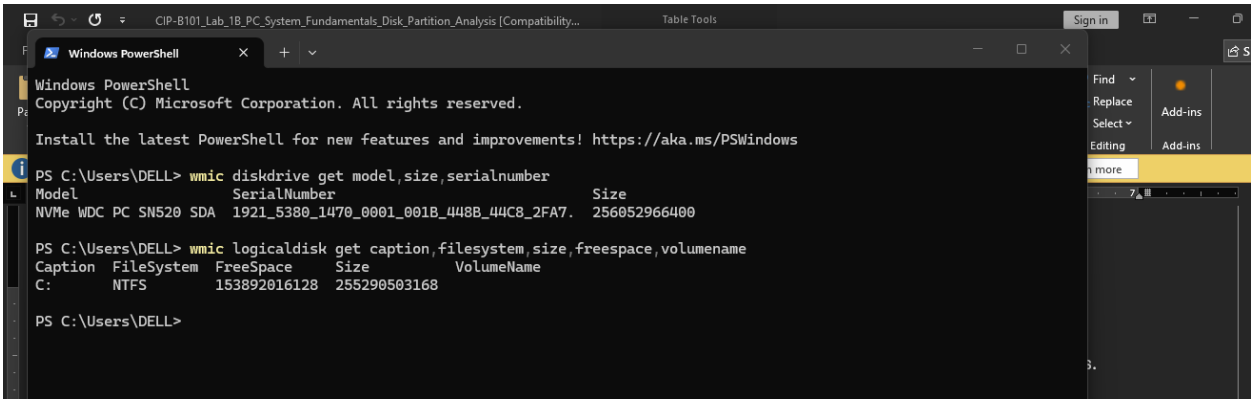
This overlap occurs due to three primary operational mechanisms:

1. The Interaction Pipeline (Data Flow). Using the framework User Activities = Applications + Operating System + Disk/Memory, we can trace how a single action replicates data.
2. Redundancy, Caching, and Optimization, Operating systems and applications constantly copy data between layers to improve performance and protect against crashes.
   RAM to Disk: When RAM fills up, the OS moves data into virtual memory swap files (pagefile.sys or swapfile) on the physical Disk. This mirrors highly volatile memory evidence onto persistent storage.
   OS Caching: The OS builds shortcuts like Prefetch files or Shellbags on the Disk to launch applications faster. This ensures that even if an application log is cleared, the OS layer preserves proof that the application ran.
3. **Corroboration and Anti-Forensics Verification,** From an investigative standpoint, multi-layer evidence acts as a security system against tampering. If a suspect attempts to cover their tracks by deleting browser history (Application layer), the OS layer may still hold a Prefetch file showing the browser was executed, and the File System layer will contain master file table records or unallocated space containing the deleted database blocks.

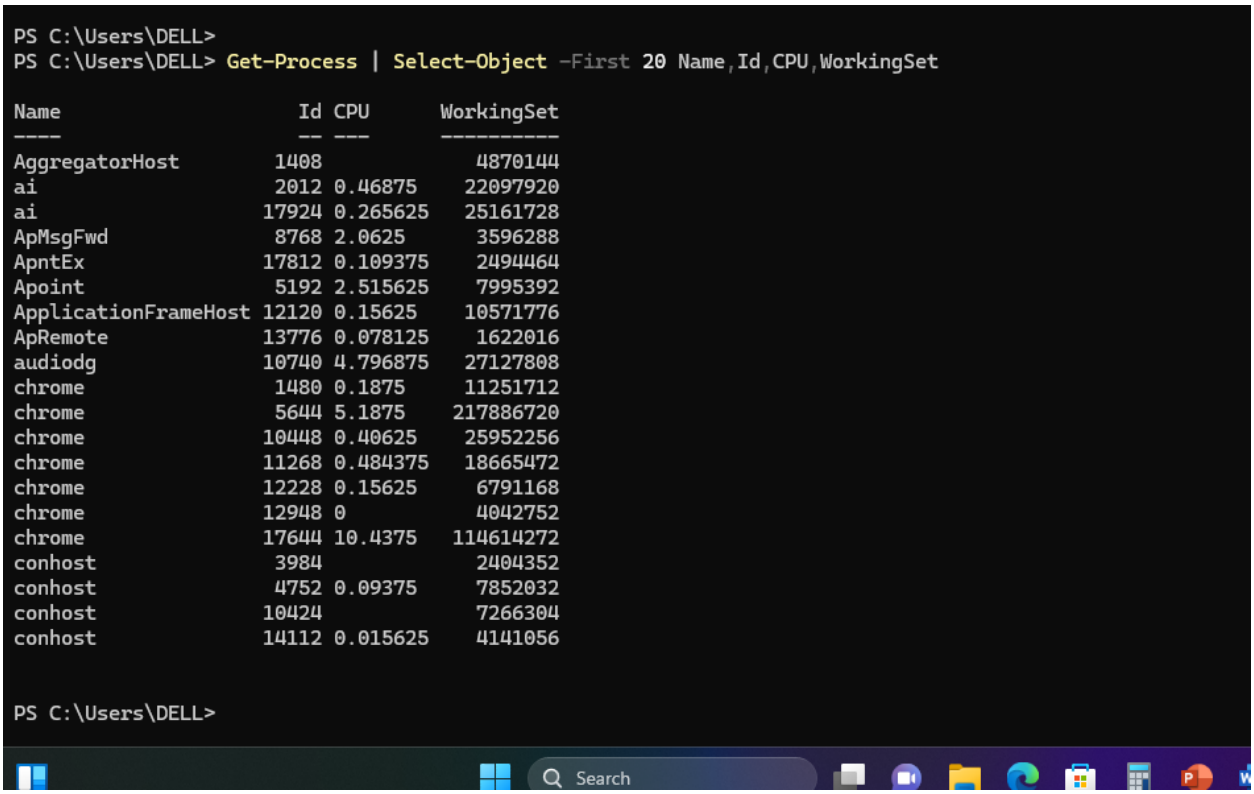
Part B - Program, Process and Thread Investigation

B1. Observe running processes

- Open Task Manager on Windows, or use Command Prompt/PowerShell.



- Select three processes and record their process names, PID values, and likely purpose.



B2. Explain the difference

1. In your report, explain the difference between a program, a process, and a thread in your own words.

   A **program** is a passive, inactive file stored on your computer's hard drive. It is nothing more than a set of written instructions and code waiting to be used. And the

   A **process** is a program that is actively running on the system.

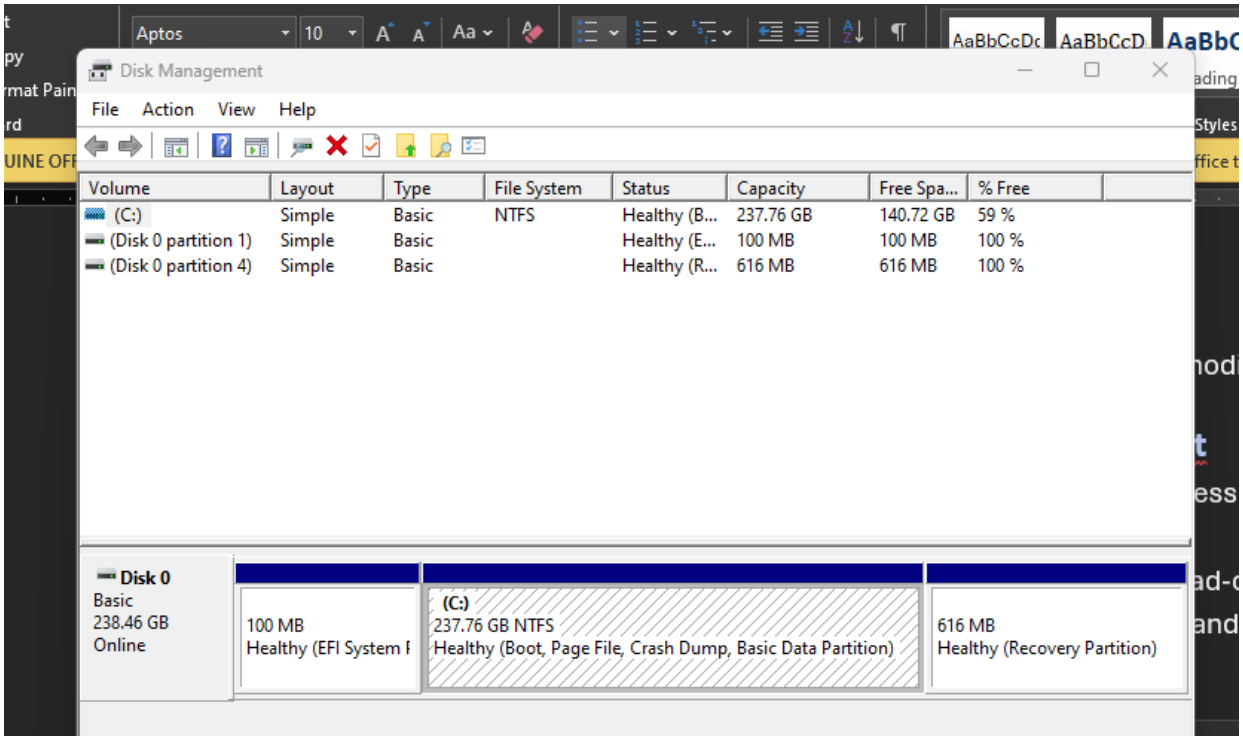
2. Explain why process information may be useful during malware analysis or live forensic triage
   Analyzing active process information is one of the most critical steps during malware analysis and

live forensic triage. Because a program (the static file) must become a process (the active execution) to do any harm, this will help to detect, uncover, identify evasion e.t.c for the safety of any systems.

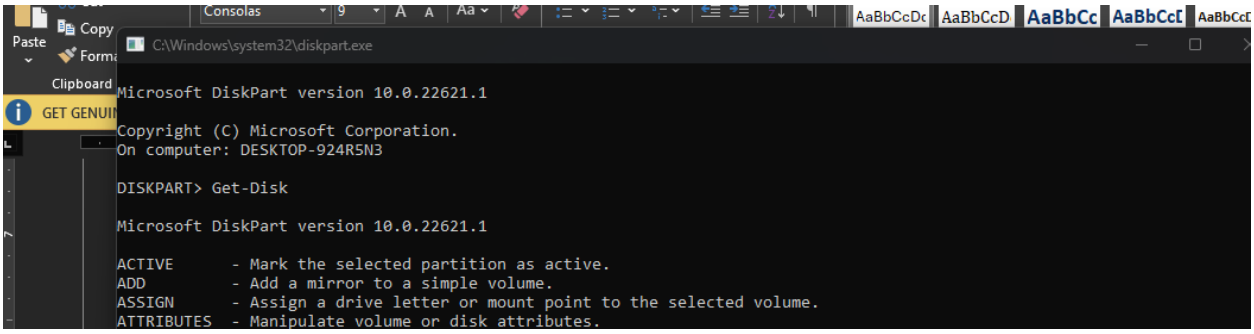
Part D - Safe Partition Table Inspection

D1. Windows method - Disk Management, PowerShell and DiskPart

1. Open Disk Management by pressing Windows + R, typing diskmgmt.msc, and pressing Enter.

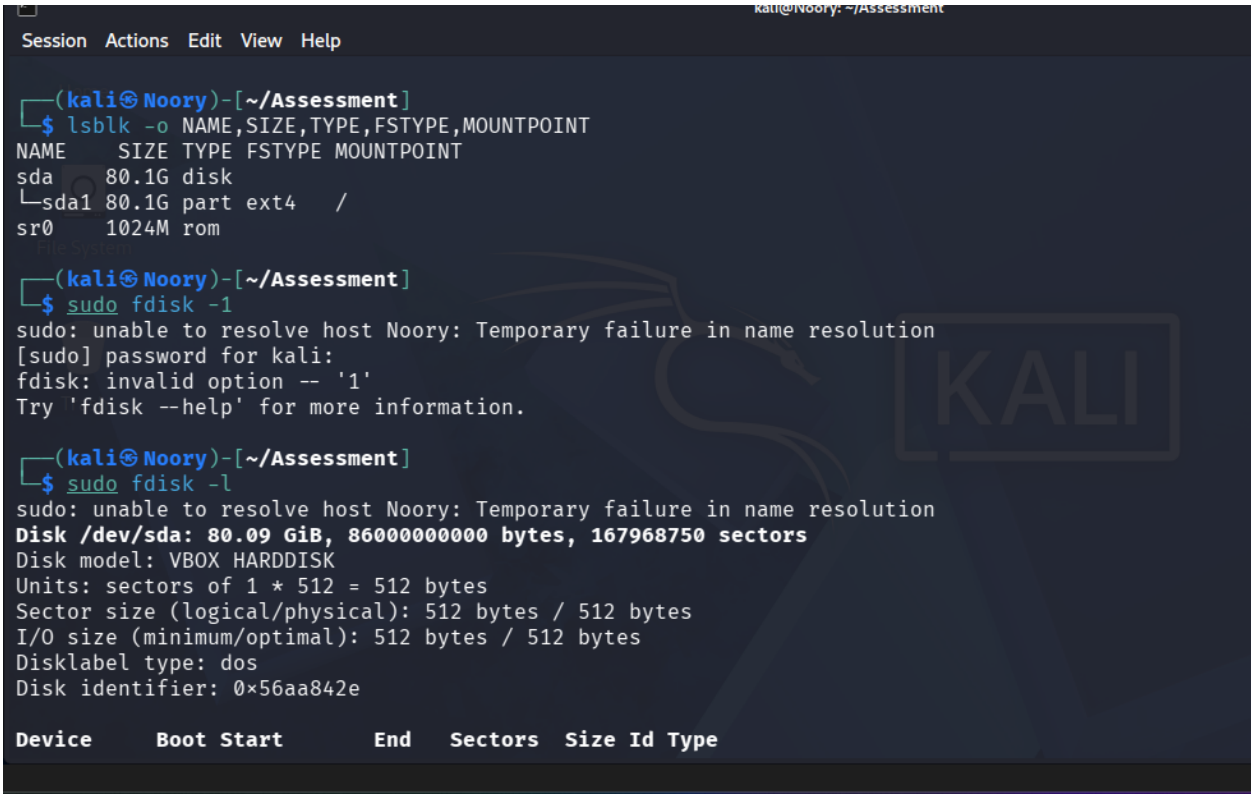


2. Open PowerShell as normal user or administrator if required and run the safe read-only commands below

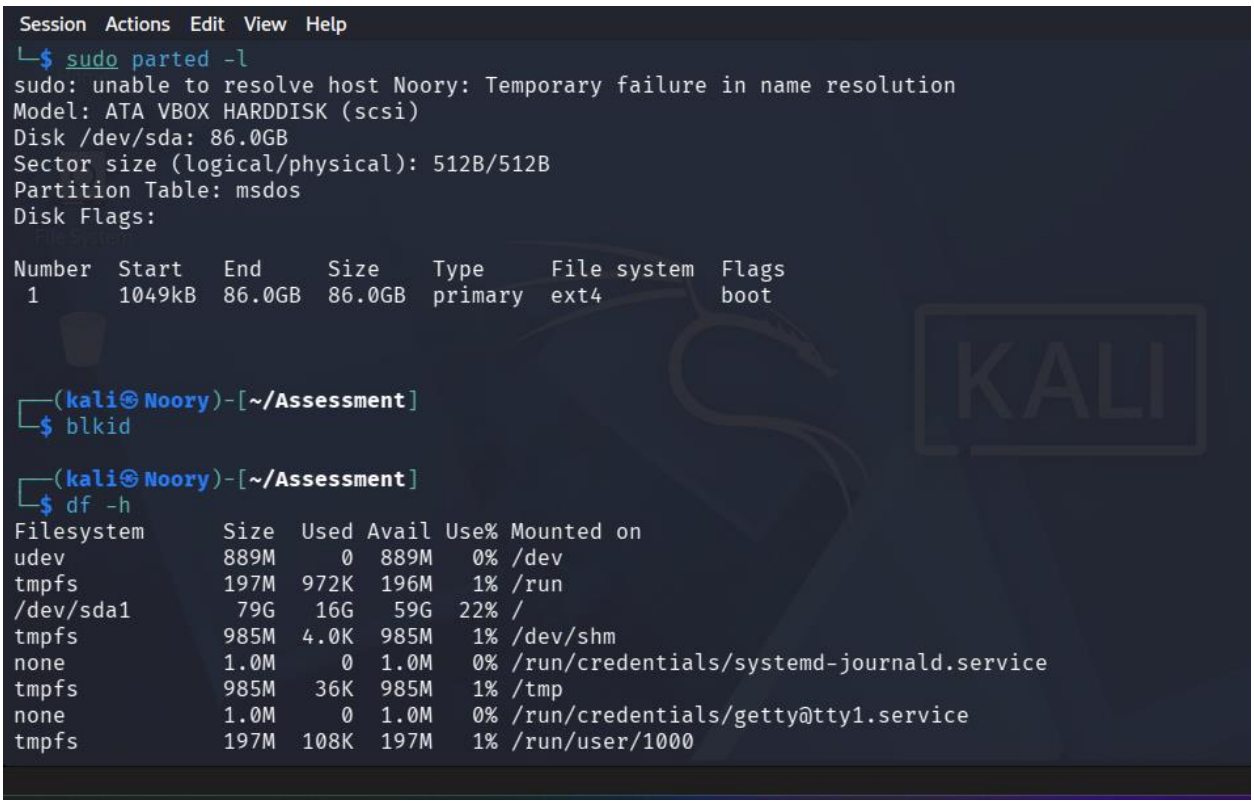


D2. Linux/Kali method - read-only inspection

The first and the second screenshots will show disk, partition, size, and file system information.



The second screenshot



Part E - MBR, GPT and Boot Process Explanation

E1. Explain the boot process

Simple flow of how a PC boots.

Step-by-Step Breakdown

1. **Power-On (The Spark)**

   When ypress the power button, the Power Supply Unit (PSU) sends electricity to the motherboard and components. Once the power stabilizes, the PSU sends a **"Power Good" signal** to the CPU, telling it that it is safe to turn on and start executing instructions.

2. **POST (Power-On Self-Test)**

   The CPU wakes up and immediately runs the **POST**, a built-in diagnostic routine. The POST quickly checks the computer's vital hardware components—such as the RAM, video card, keyboard, and storage drives—to ensure they are connected and functioning properly. If something is broken, the PC will emit a series of warning "beep codes" or flash debug lights.

3. **BIOS/UEFI Initialization**

   Once the POST passes, control is handed over to the motherboard's built-in firmware: either the legacy **BIOS** (Basic Input/Output System) or the modern **UEFI** (Unified Extensible Firmware Interface). This firmware initializes the essential hardware so the system can read storage disks and display text on the screen.

4. **CMOS / Firmware Settings Check**

   The BIOS/UEFI reads the system configuration data stored in the **CMOS memory** (a small chip kept alive by a coin-cell battery). This tells the firmware your specific hardware preferences, the system time, and crucially, the **Boot Priority Order** (e.g., whether to boot from the internal SSD first, a USB drive, or an optical disk).

5. **Reading MBR/GPT or EFI Boot Records**

   The firmware looks at the top-priority storage drive to find out *how* to start the operating system based on its formatting style:

   - **Legacy (BIOS + MBR):** The BIOS looks at the very first sector of the hard drive, called the **Master Boot Record (MBR)**, to find the active partition where the operating system resides.
   - **Modern (UEFI + GPT):** The UEFI looks at a **GUID Partition Table (GPT)** to locate a special, hidden partition known as the **EFI System Partition (ESP)**. This partition contains standard boot files, eliminating the need for a single, fragile MBR sector.

6. **Executing the Boot Loader / Manager**

   Once the boot sector or partition is found, the system loads the **Boot Loader** (such as Windows Boot Manager for Windows, or GRUB for Linux) into the computer's RAM. The boot loader presents the boot menu (if you have multiple operating systems installed) and prepares the environment to launch the core OS.

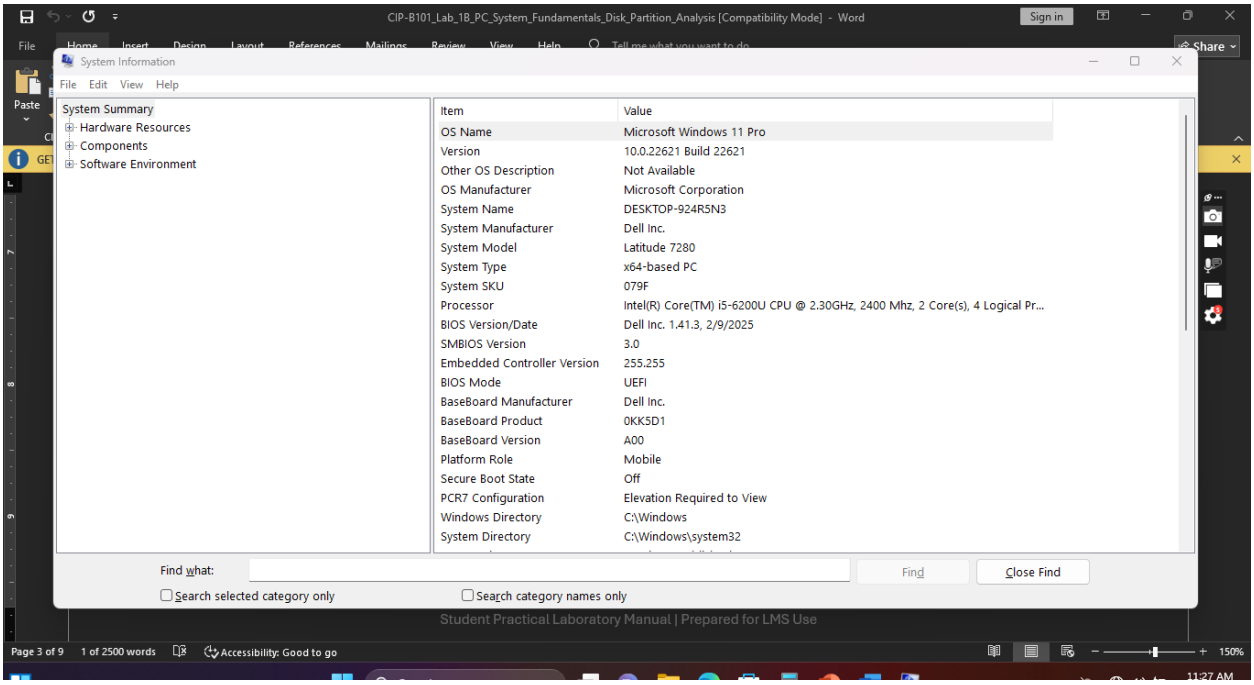
7. **Kernel Loading**

   The boot loader locates and extracts the **Operating System Kernel** (the absolute core program of the OS) from the storage drive and loads it into the RAM. It also loads essential low-level drivers that the kernel needs to talk to your specific hardware.

8. **Operating System Control**

   The boot loader hands complete execution control over to the **Kernel**. The kernel initializes all background system services, mounts the main user interface (like the Windows Desktop or macOS login screen), and sits back to manage the system while waiting for user inputs.

My system has UEFI based on system



E2. Compare MBR and GPT

To compare between the two MBR and GPT

1. Disk Size and Partition Limits

- **MBR** uses a 32-bit table to keep track of sectors, which caps its maximum addressing space at **2 TB**. If you plug a 4 TB drive into an MBR system, the computer will see the remaining 2 TB as unallocated space that cannot be used. It is also limited to **4 primary partitions**.
- **GPT** uses 64-bit logical block addressing, pushing the theoretical limit to **9.4 Zettabytes** (over 9 billion gigabytes). It handles up to **128 separate partitions** directly without needing complex workarounds like extended or logical drives.

2. Reliability and Redundancy

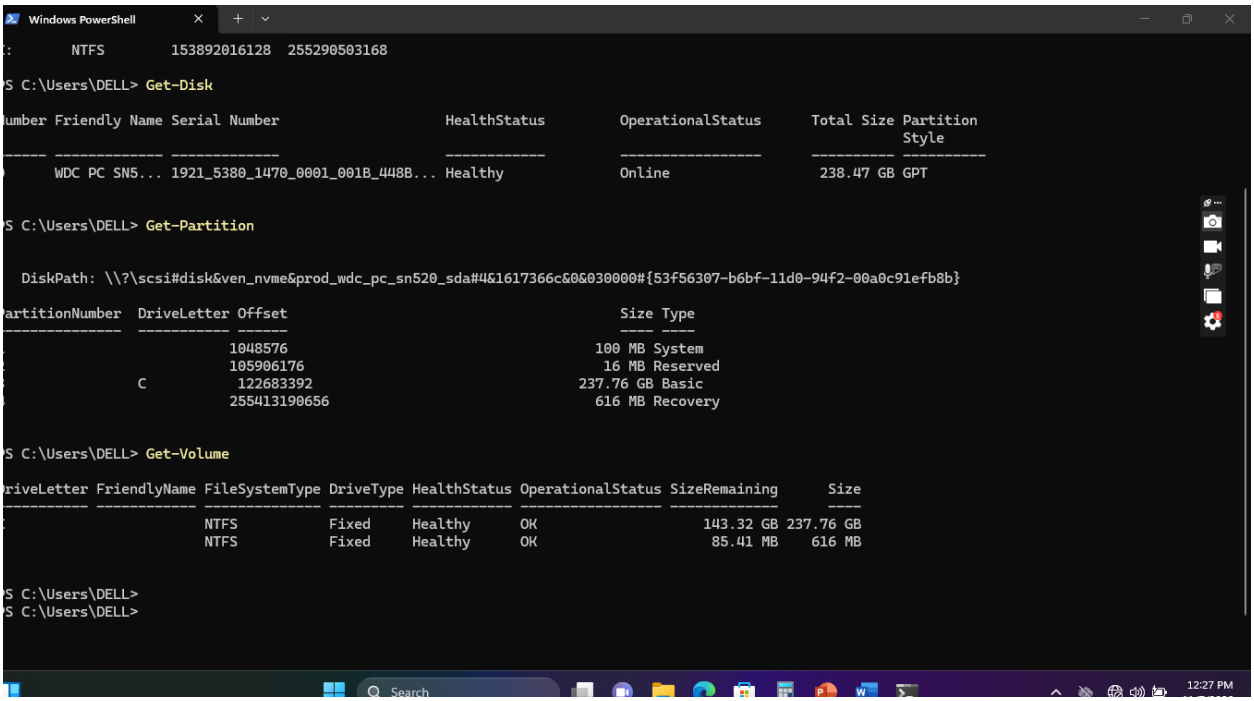
- **MBR** stores all partition and boot data in the **very first sector** of the drive. If this specific sector becomes corrupted, suffers a bad sector, or is overwritten by malware, the entire drive becomes unreadable and the PC will fail to boot.
- **GPT** distributes this risk by writing the partition table at the **beginning of the drive and creating a mirror copy at the very end of the drive**. If the primary table is damaged, the system automatically restores it using the backup copy. It also uses Cyclic Redundancy Checks (CRC32) to actively monitor the table for corruption.

3. Forensic & Security Implications

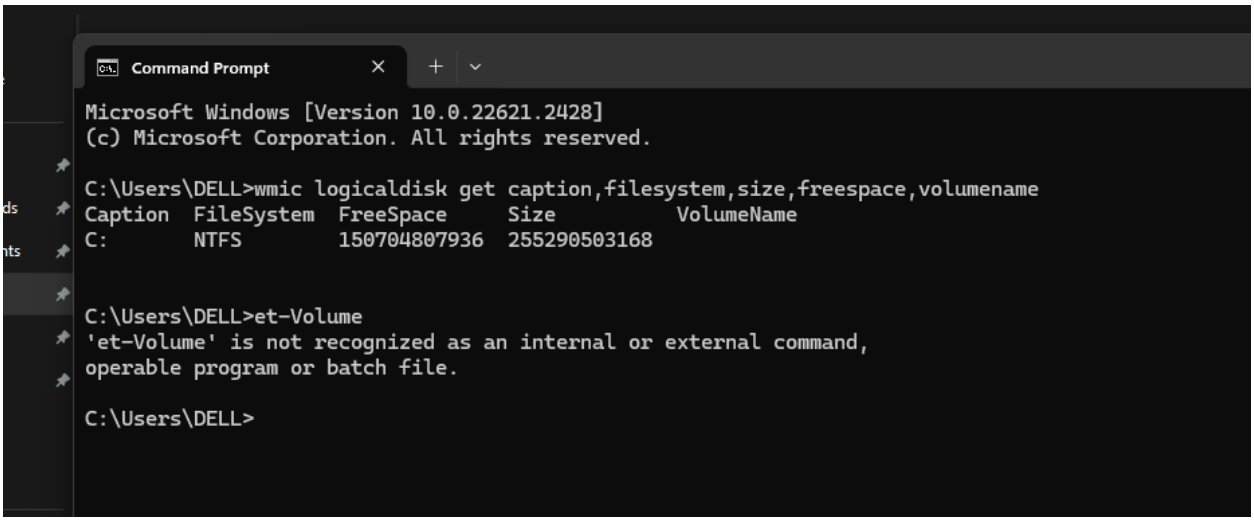
- **Malware Vulnerability:** Because MBR executes code directly from the boot sector before the operating system starts, it is a frequent target for a type of stealthy malware called a **Bootkit**.
- **Protective MBR:** GPT drives include a "Protective MBR" sector at the very beginning. If an old legacy tool or OS attempts to read a GPT disk, it sees it as a single, giant, protected MBR partition rather than an empty disk. This prevents older disk utilities from accidentally overwriting your GPT partitions.

Part F - File System Identification and Interpretation

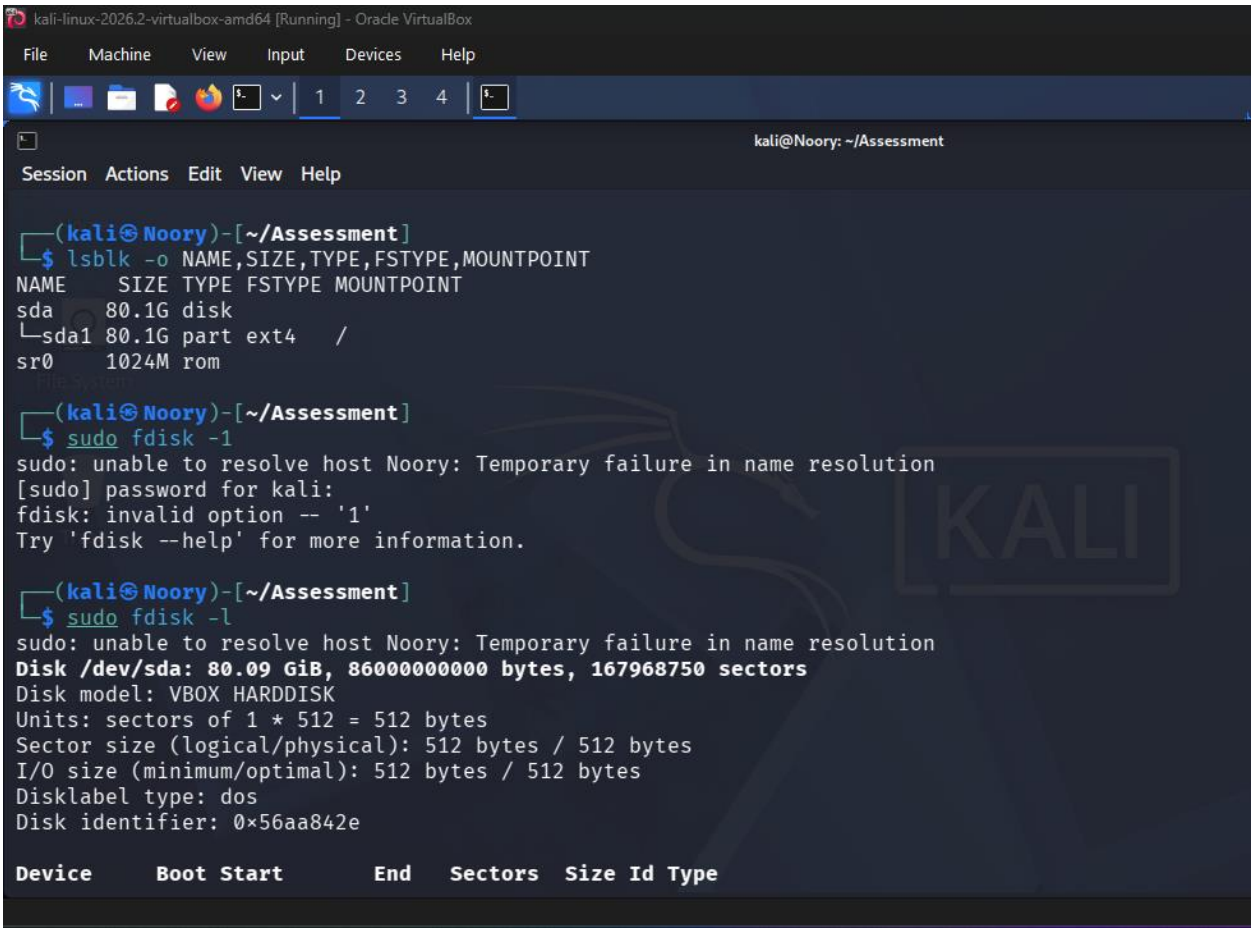
- Identify the file system used by at least two volumes or partitions on your system.



- For Windows systems, identify whether volumes use NTFS, FAT/FAT32, exFAT, or another file system.



- For Linux systems, identify whether partitions use ext4, XFS, btrfs, swap, or another file system.



- In your report, explain why a file system is important in digital forensics.
  A **file system** is one of the most critical components in digital forensics because it dictates the structural rules for how data is stored, organized, and retrieved on a physical storage device